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Slot-B

EXPERIMENT-1

1. Perform basic Image Handling and processing operations on the image.

AIM:- Read an image in python and Convert an Image to Grayscale

Algorithm:-

1. Start the program.
2. Import the required library (cv2).
3. Read the input image using the cv2.imread() function.
4. Check whether the image is loaded successfully.
5. Convert the color image to grayscale using cv2.cvtColor(image, cv2.COLOR_BGR2GRAY).
6. Display the original image.
7. Display the grayscale image.
8. Wait for a key press using cv2.waitKey(0).
9. Close all image windows using cv2.destroyAllWindows().
10. Stop the program.

Output :-



Result:- The program is executed successfully.

EXPERIMENT-2

2) Perform basic Image Handling and processing operations on the image

AIM:- Read an image in python and Convert an Image to Blur using GaussianBlur.

Algorithm:-

1. Start the program.
2. Import the required library (cv2).
3. Read the input image using the cv2.imread() function.
4. Check whether the image is loaded successfully.
5. Apply Gaussian Blur to the image using cv2.GaussianBlur(image, (5,5), 0).
6. Display the original image.
7. Display the blurred image.
8. Wait for a key press using cv2.waitKey(0).
9. Close all image windows using cv2.destroyAllWindows().
10. Stop the program.

Output:-



Result:- The program is executed successfully.

EXPERIMENT-3

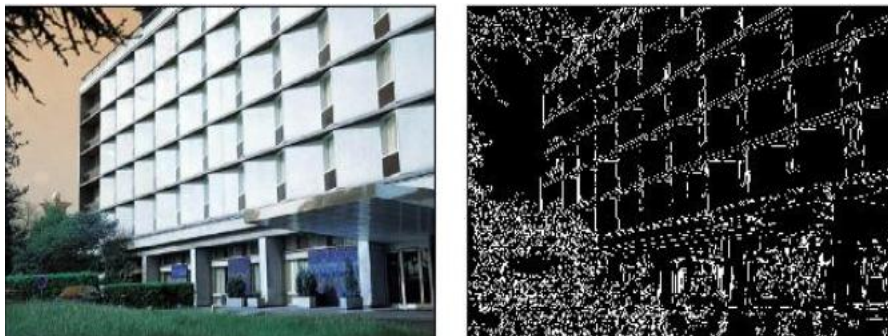
3) Perform basic Image Handling and processing operations on the image

AIM:- Read an image in python and Convert an Image to show outline using Canny function.

Algorithm:-

1. Start the program.
2. Import the required library (cv2).
3. Read the input image using the cv2.imread() function.
4. Check whether the image is loaded successfully.
5. Convert the image to grayscale using cv2.cvtColor().
6. Apply the Canny edge detection algorithm using cv2.Canny(gray_image, 100, 200).
7. Display the original image.
8. Display the image showing the detected outlines (edges).
9. Wait for a key press using cv2.waitKey(0).
10. Close all image windows using cv2.destroyAllWindows().
11. Stop the program.

Output:-



Result:- The program is executed successfully.

EXPERIMENT-4

4) Perform basic Image Handling and processing operations on the image

AIM:- Read an image in python and Dilate an Image using Dilate function.

Algorithm:-

1. Start the program.
2. Import the required library (cv2) and numpy.
3. Read the input image using the cv2.imread() function.
4. Check whether the image is loaded successfully.
5. Create a kernel (structuring element) using numpy.ones().
6. Apply dilation to the image using cv2.dilate(image, kernel, iterations=1).
7. Display the original image.
8. Display the dilated image.
9. Wait for a key press using cv2.waitKey(0).
10. Close all image windows using cv2.destroyAllWindows().
11. Stop the program.

Output:-



Result:- The program is executed successfully.